

Induction Motor Fault Diagnosis using Convolutional Neural Networks: A Physics-Aware Approach

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Abstract—Induction motors are the primary drivers of industrial machinery, making their reliability critical for production efficiency and safety. This paper presents a robust, automated fault diagnosis system based on Motor Current Signature Analysis (MCSA) and Deep Learning. We developed high-fidelity MATLAB Simulink models for a Squirrel Cage Induction Motor (SCIM) under six operating conditions: Healthy, Broken Rotor Bar, Eccentricity, Bearing Fault, Inter-Turn Stator Short, and Supply Voltage Imbalance. To address the challenge of distinguishing subtle fault signatures under grid instability, we implemented a physics-guided signal processing pipeline utilizing high-resolution Short-Time Fourier Transforms (STFT) converted into RGB spectrograms. A Convolutional Neural Network (CNN) was trained on 3,000 randomized simulations, achieving a test accuracy of 98.1% and demonstrating robust performance against variable load torque and supply frequency drift.

I. INTRODUCTION

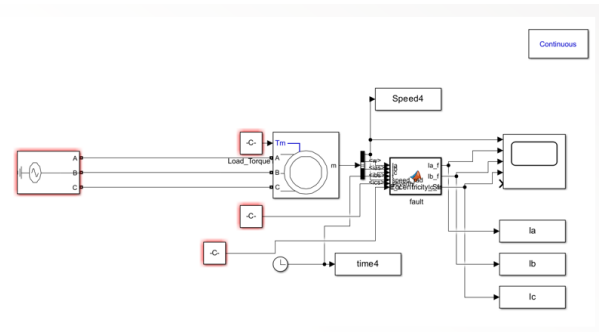
Induction motors drive a significant portion of industrial power, serving as the backbone for pumps, fans, compressors, and manufacturing conveyors. Due to their widespread use, unexpected failures can cause huge safety hazards and substantial production losses. Consequently, effective condition monitoring is essential to transition from reactive maintenance to predictive maintenance strategies. Traditional fault diagnosis relies heavily on expert manual analysis of vibration or current data. This approach is costly, time-consuming, and prone to human error, particularly when fault signatures are masked by operational noise. Furthermore, existing automated systems often fail when subjected to realistic grid fluctuations, such as varying voltage or frequency drift.

II. SYSTEM METHODOLOGY AND FAULT MODELLING

The proposed system follows a workflow of physics-based data generation, signal preprocessing, and deep learning classification.

A. Simulation Setup

We utilized MATLAB Simulink to generate a comprehensive dataset for a 7.5 kW asynchronous machine. To ensure the model is robust to real-world variability, we randomized physical parameters for every simulation run while adding 1% white noise. The dataset consists of 500 simulations for each of the six classes.



The operational parameters were randomized as follows:

- **Supply Frequency:** Randomized between 49.5 Hz and 50.5 Hz to simulate grid instability.
- **Load Torque:** Randomized between 10 Nm and 60 Nm to ensure variable slip. This variation is crucial as it shifts the frequency location of slip-dependent faults, preventing the neural network from memorizing fixed pixel locations.
- **Input Voltage:** Randomized between 350 V and 450 V to mimic voltage sags and swells common in industrial environments.
- **Fault Severity:** Randomized custom ranges for every fault to cover minor to severe fault conditions, reflecting real life scenarios.

B. Electromechanical Fault Modelling

The core contribution of this work is the rigorous physical Modelling of five specific motor faults.

1. **Broken Rotor Bar (BRB):** This fault introduces an asymmetry in the rotor cage, preventing current flow in the broken bar. This was Modelled via Amplitude Modulation (AM) of the stator current, creating characteristic sidebands around the fundamental frequency at 50Hz, where is the slip.
2. **Eccentricity Fault:** Modelled as mixed eccentricity (a combination of static and dynamic air-gap irregularities). This creates a modulation of the air-gap flux density, resulting in sidebands $fs \pm 25\text{Hz}$, where is the rotor mechanical frequency.
3. **Bearing Fault:** Bearing defects create mechanical vibrations that modulate the air gap. This was Modelled as a series of high-frequency impulses

and modulations based on the Ball Pass Frequency of the Outer Race (BPFO).

4. **Inter-Turn Stator Winding Short:** This fault represents a short circuit between turns in the same phase winding. It was Modelled by injecting odd harmonics (3rd, 5th, 7th) into the current signal to simulate the distortion of the magnetic flux distribution caused by the short.
5. **Supply Voltage Imbalance:** This condition creates negative sequence currents that cause overheating. It was Modelled by reducing the amplitude of Phase A voltage by 5%–15%, creating a distinct asymmetry in the three-phase current magnitudes.

III. Signal Processing Pipeline

To extract actionable features from the raw current data, a specialized preprocessing pipeline was implemented.

A. Sampling and Decimation

Each simulation generated 5 seconds of data at a 10 kHz sampling frequency. To focus on the relevant spectral range (0–400 Hz) where electromechanical fault signatures reside, the raw samples were decimated to 1 kHz. The first 1

second of data was trimmed to remove initial startup transients and inrush current spikes.

B. High-Resolution Spectral Analysis

A critical challenge in MCSA is resolving the Broken Rotor Bar sidebands, which can be less than 1 Hz away from the 50 Hz fundamental frequency at low slip. To address this, we implemented a High-Resolution Short-Time Fourier Transform (STFT) with a frequency resolution of 0.33 Hz. A Chebyshev window was applied to minimize spectral leakage, enhancing the visibility of weak fault signals adjacent to the strong fundamental peak.

C. RGB Spectrogram Generation

The processed 3-phase currents were mapped to the Red, Green, and Blue channels of an image. This encoding allows the Convolutional Neural Network (CNN) to utilize inter-phase correlation features, effectively visualizing imbalances and phase shifts.

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IV. EXPERIMENTAL RESULTS AND DISCUSSION

A. Spectral Feature Verification

Before training, the simulated data was validated against theoretical expectations to ensure physical accuracy.

Broken Rotor Bar: Visible "Trident" pattern with sidebands around the fundamental, confirming the slip-dependent modulation.

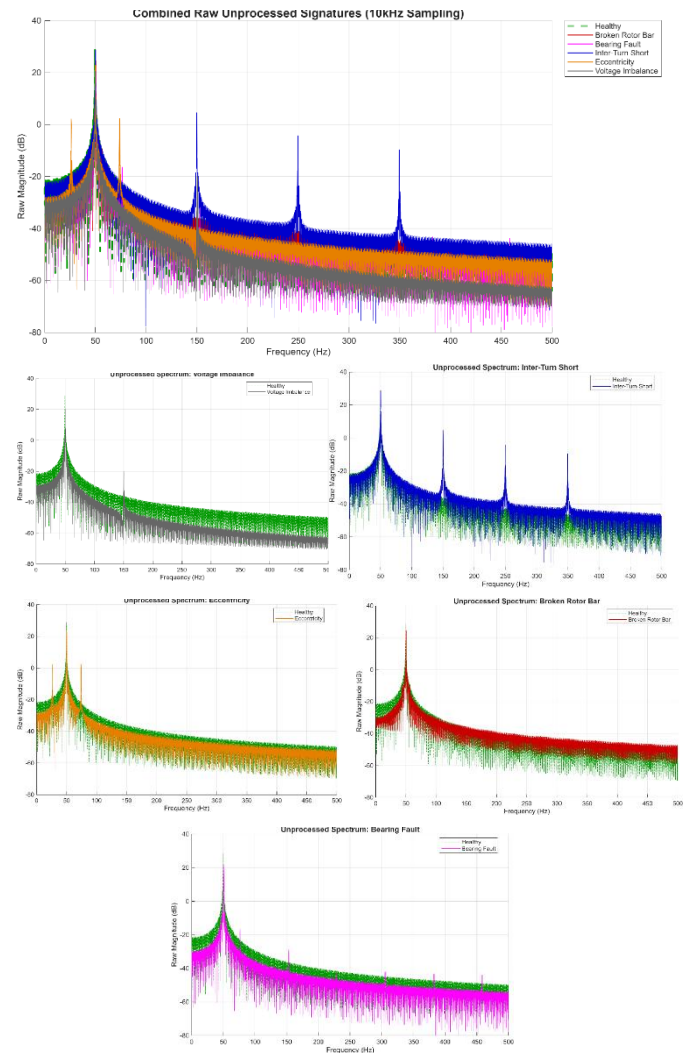
Eccentricity: Distinct sidebands observed at $\pm 25\text{Hz}$ from fundamental, consistent with mixed eccentricity theory.

Bearing Fault: A strong characteristic peak was detected at 75 Hz, corresponding to the BPFO modulation.

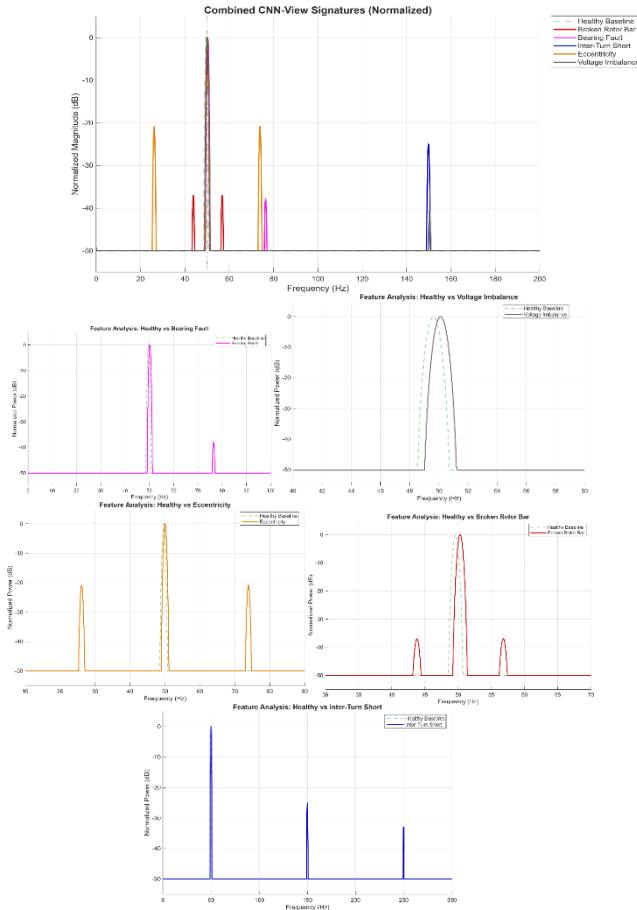
Inter-Turn Short: Strong peaks detected at 3rd harmonic and 5th harmonic.

Voltage Imbalance: Evidenced by a % difference in amplitude, representing a shift in average power distribution.

Before Data Processing:



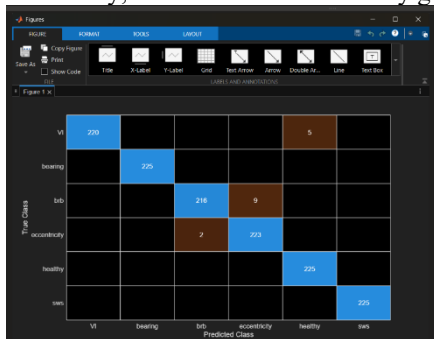
After Data Processing:



B. Classification Performance

The CNN model was trained on the dataset of 3,000 simulations. The training process utilized a standard 70-15-15 split for training, validation, and testing.

- **Overall Accuracy:** The model achieved a test accuracy of 98.1%.
- **Fault-Specific Performance:** The model exhibited high precision across most classes. However, it showed the lowest accuracy (96%) in identifying Broken Rotor Bars. This is attributed to the very small sideband amplitudes in cases of low fault severity, which can be obscured by grid noise.



Fault	Prediction Accuracy	Remarks
Broken Rotor Bar (BRB)	96%	Sidebands close to Fundamental
Bearing Fault	100%	Very distinct sidebands
Inter turn Short Circuit	100%	Distinct sidebands
Eccentricity Fault	99.11%	Similar sidebands to BRB
Voltage Imbalance	97.78%	Similar average power to healthy

C. Challenges and Solutions

1. **Resolution Limits:** The CNN initially confused healthy motors with Broken Rotor Bar faults. This was resolved by increasing the STFT window size to 3 seconds and applying Chebyshev windowing, which visually separated the fault sidebands.
2. **Grid Frequency Drift:** The model failed when the grid frequency drifted away from exactly 50 Hz. We implemented Vertical Translation (Data Augmentation) during training, teaching the CNN to learn the *relative* fault pattern rather than memorizing exact pixel positions.

V. CONCLUSION AND FUTURE WORK

This study demonstrates that Deep Learning, specifically CNNs trained on physics-aware spectrograms, can accurately diagnose complex electromechanical faults in induction motors. The use of RGB spectrograms proved highly effective in differentiating between phase-related faults like voltage imbalance and harmonic-related faults like inter-turn shorts.

Limitations: The current model relies on simulation data due to hardware resource constraints. While noise and drift were modelled, real industrial environments may contain impulsive interference or extra noise not captured in the training.

Future Work: Future development will focus on deploying the trained model on an embedded system, such as a Raspberry Pi, equipped with current sensors for real-time monitoring on a physical motor test rig.

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